

INTRODUCTION

SESSION 1
COMP6140001 – DATA MINING

SUBJECT MATTER EXPERT
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LEARNING OBJECTIVE

L01 : Explain concept of data, data warehouse, data preprocessing, and data mining



OUTLINE

- Data mining definition
- Knowledge discovery (KDD) process
- Mining various kinds of knowledge
- Confluence of multiple disciplines
- Data mining and application

DATA MINING DEFINITION

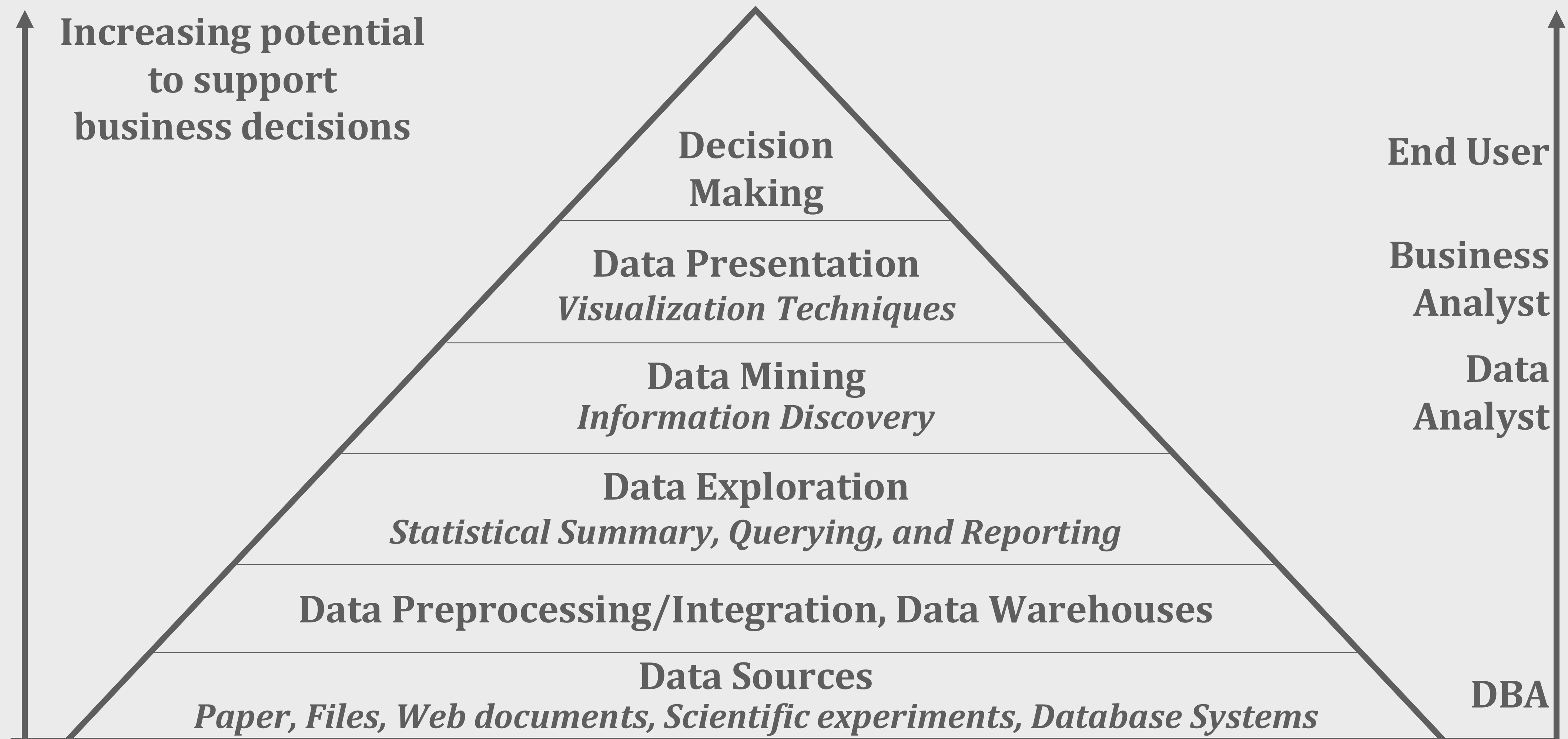


DATA MINING DEFINITION

- We are living in the information age.
- Terabytes or petabytes of data pour into our computer and various data storage devices every day.
- Data Mining is the process of discovering interesting patterns, models, and other kinds of knowledge in large data sets.
- “Mining” is a vivid term characterizing the process that finds a small set of precious nuggets from a great deal raw material.
- Other term similar meaning to data mining:
 - Knowledge mining from data
 - Knowledge discovery from data
 - Pattern discovery
 - Knowledge extraction
 - Data Analytics



DATA MINING IN BUSINESS INTELLIGENCE

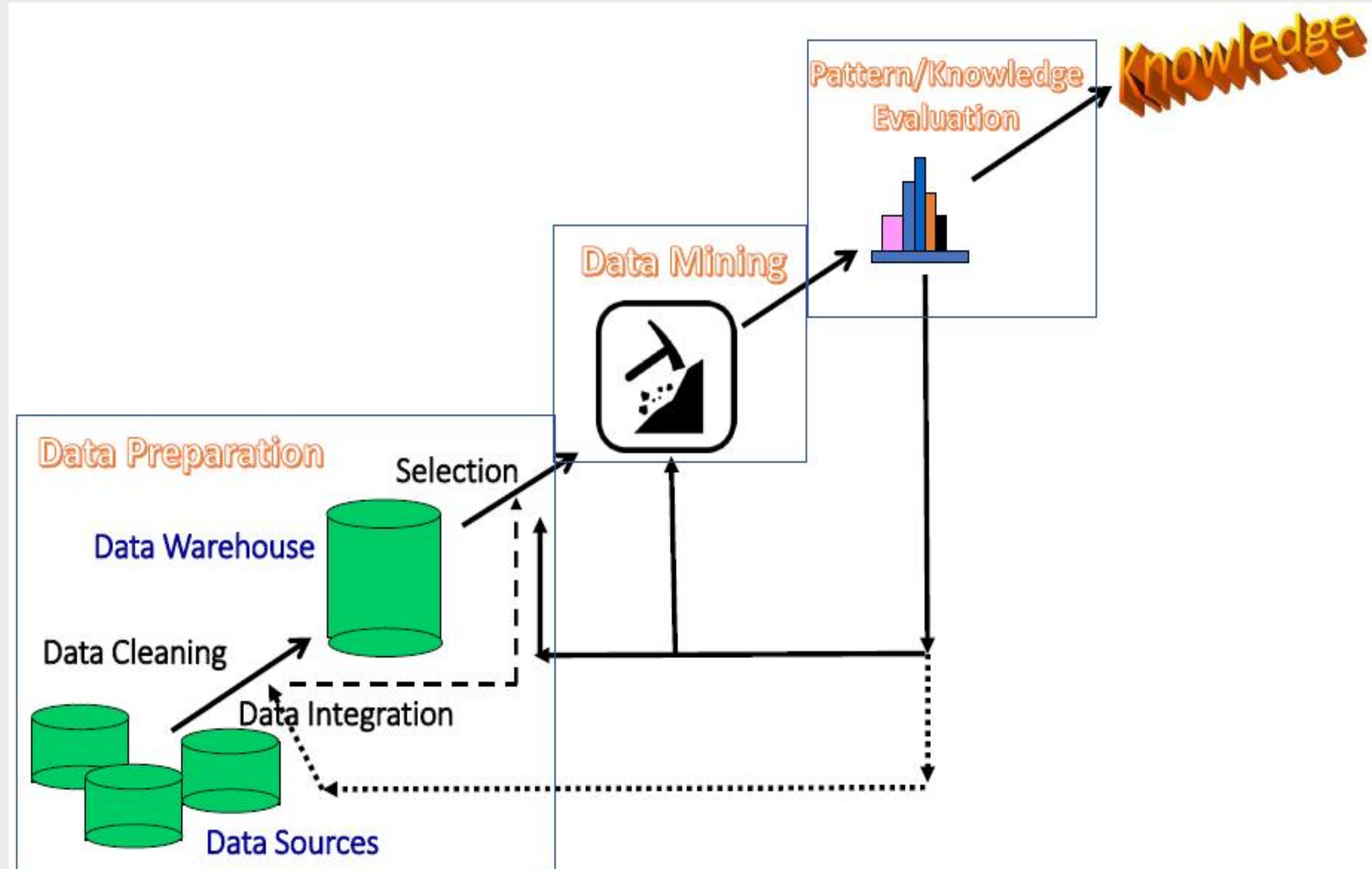


KNOWLEDGE DISCOVERY (KDD) PROCESS



KNOWLEDGE DISCOVERY (KDD) PROCESS

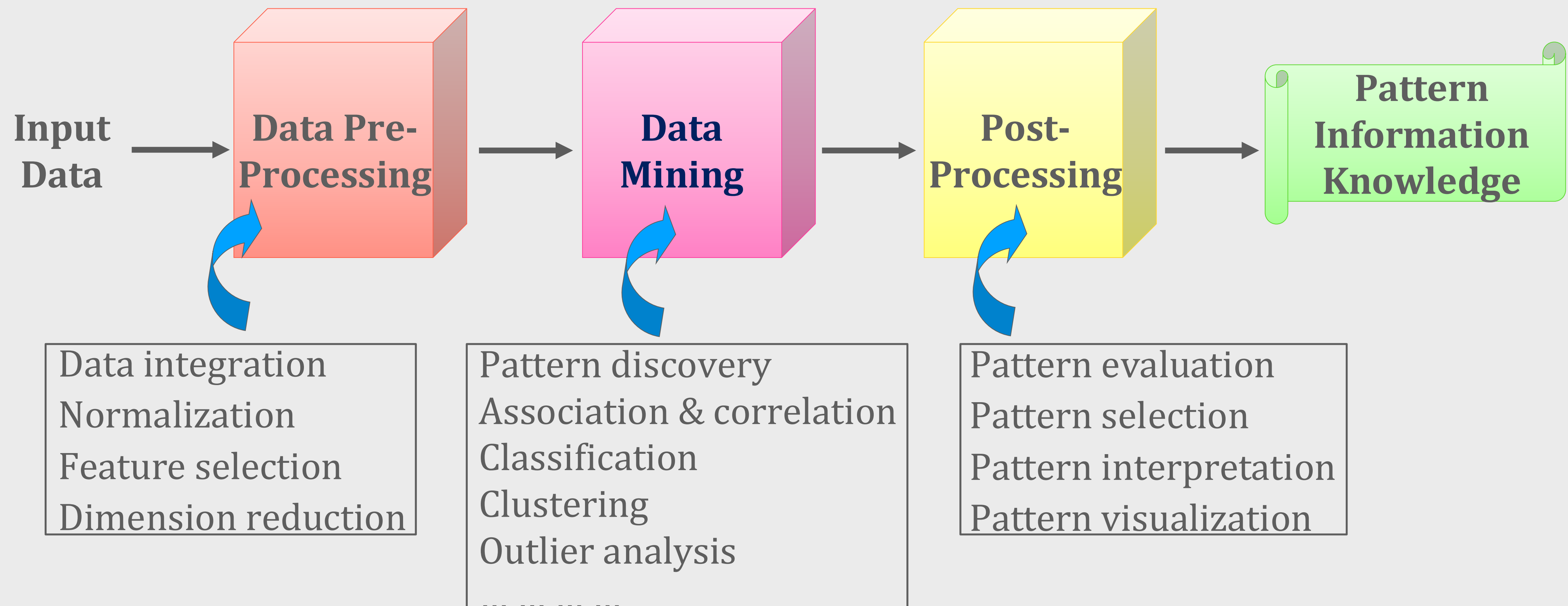
- Data preparation
 - Data cleaning
 - Data integration
 - Data transformation
 - Data selection
- Data mining
- Pattern/model evaluation
- Knowledge presentation





KNOWLEDGE DISCOVERY (KDD) PROCESS (2)

- Based on machine learning (ML) and statistics communities



MINING VARIOUS KINDS OF KNOWLEDGE



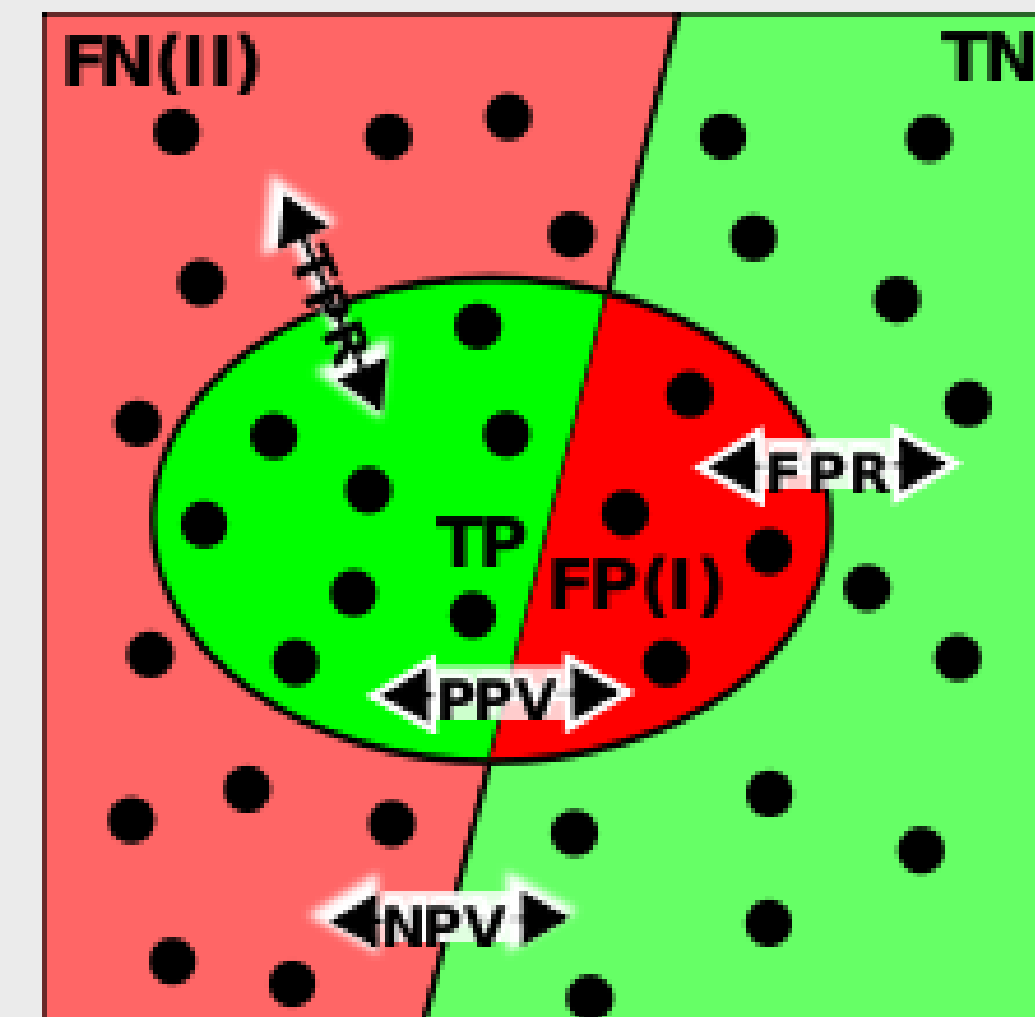
MINING VARIOUS KINDS OF KNOWLEDGE

- Different Pattern and Knowledge can be uncovered via data mining.
- In general **data mining task** can be put into two categories:
 - **Descriptive Data Mining:** Characterizes properties of the interested set of data.
 - **Predictive Data Mining:** Perform induction on the data set in order to make prediction



DATA MINING TASKS

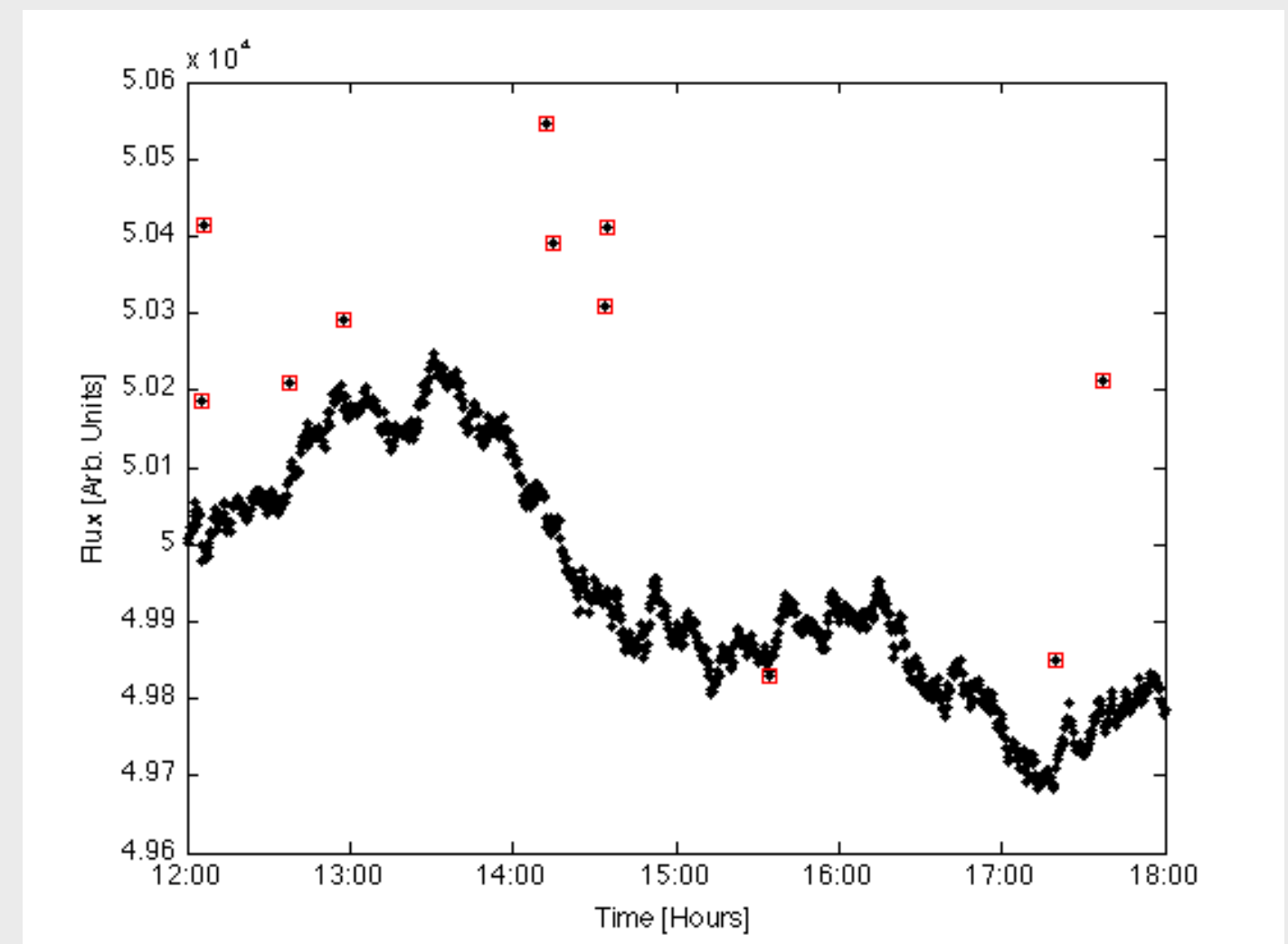
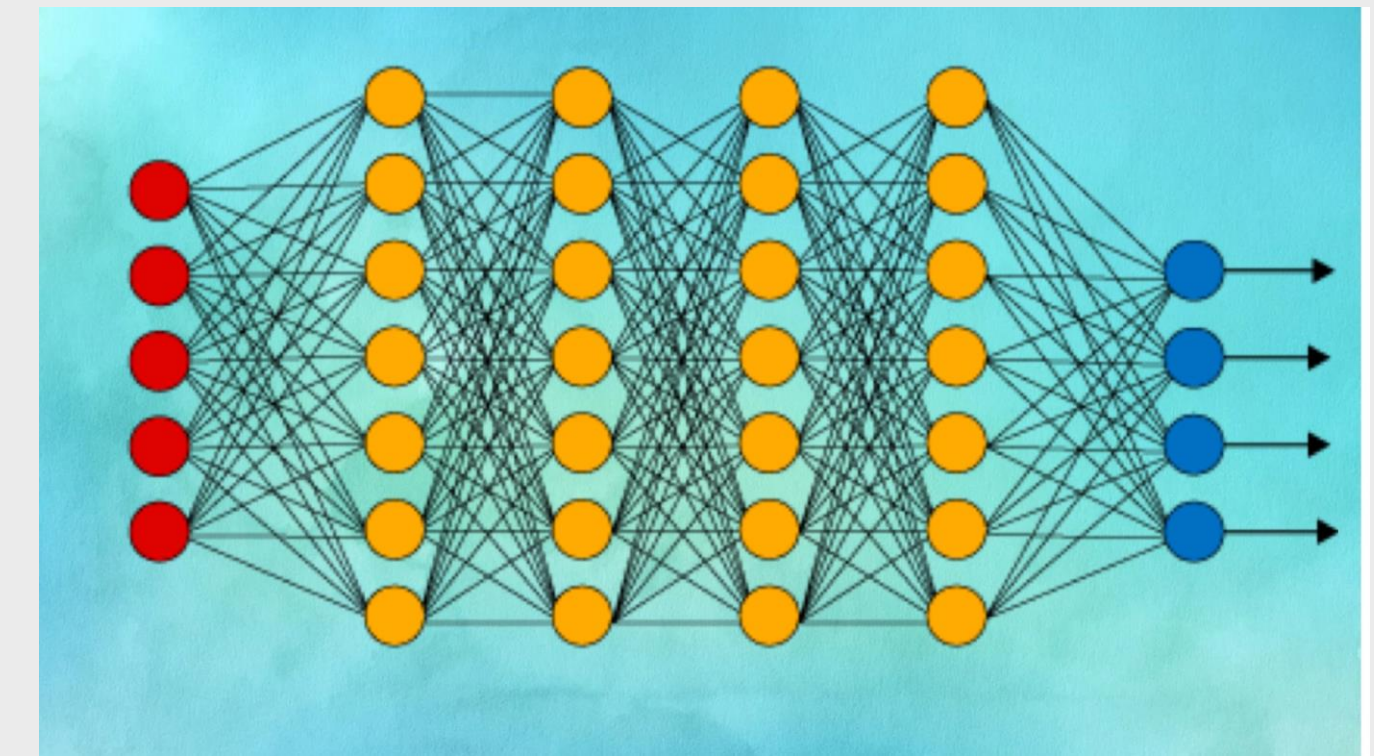
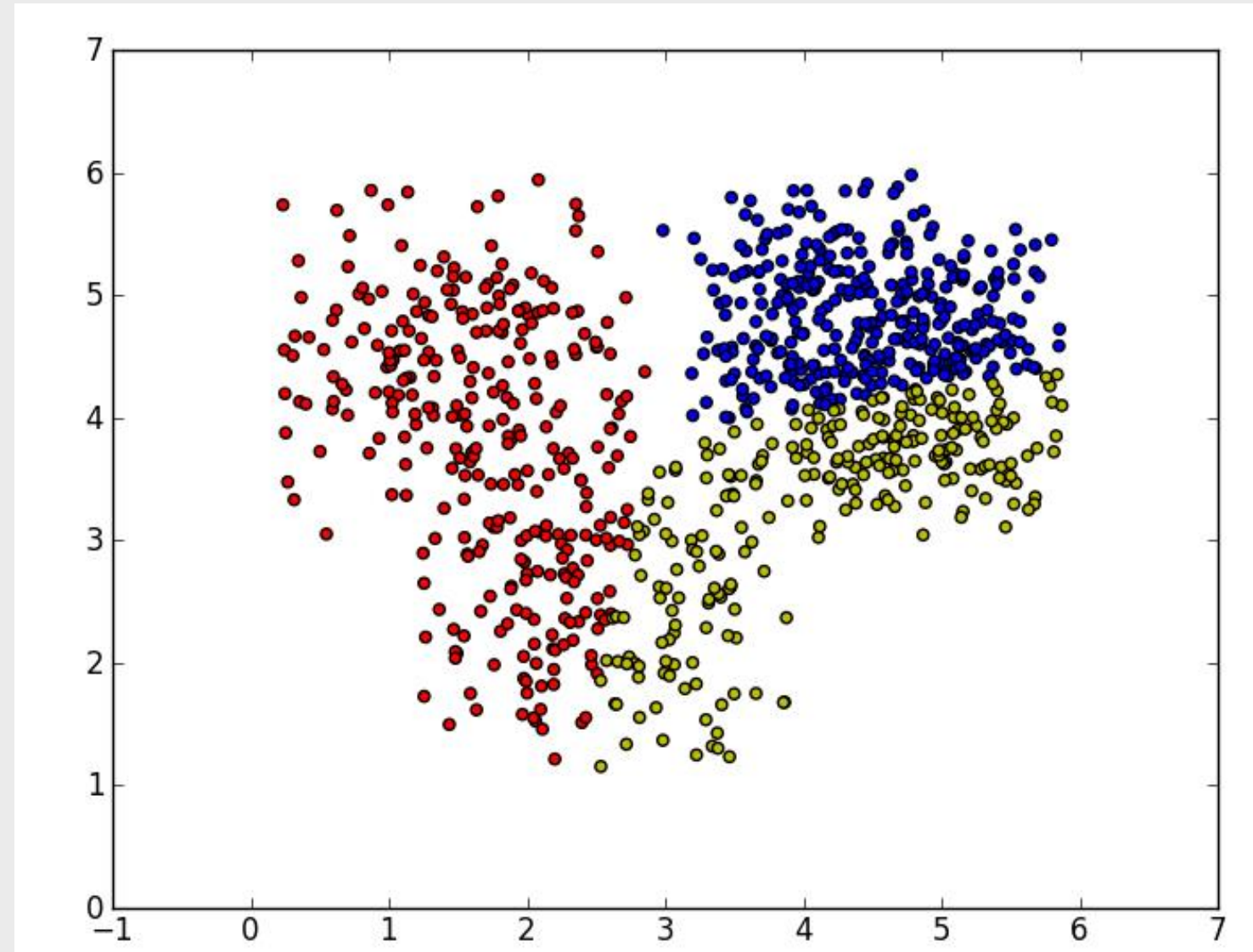
- Multidimensional Data Summarization.
Generalize, summarize, and contrast data characteristics, e.g., dry vs. wet region
- Mining Frequent Patterns, Associations, and Correlations
What items are frequently purchased together in your Walmart?
- Classification and Regression for Predictive Analysis
Describe and distinguish classes or concepts for future prediction





DATA MINING TASKS

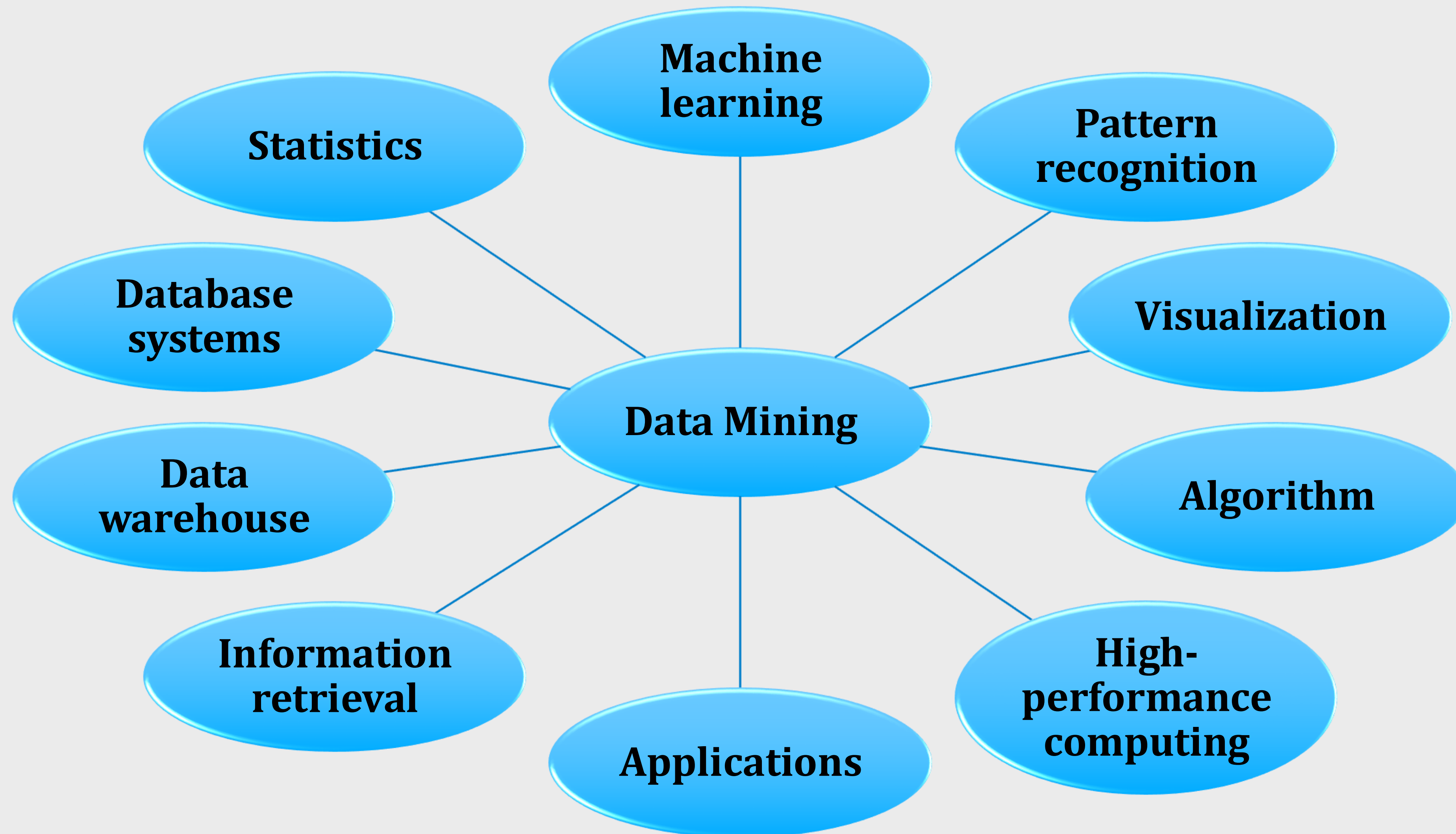
- Cluster Analysis
Group data to form new categories
- Deep Learning
Has broad applications in computer vision, natural language processing, machine translation etc.
- Outlier Analysis
A data object that does not comply with the general behavior of the data.



CONFLUENCE OF MULTIPLE DISCIPLINES



CONFLUENCE OF DISCIPLINES IN DATA MINING





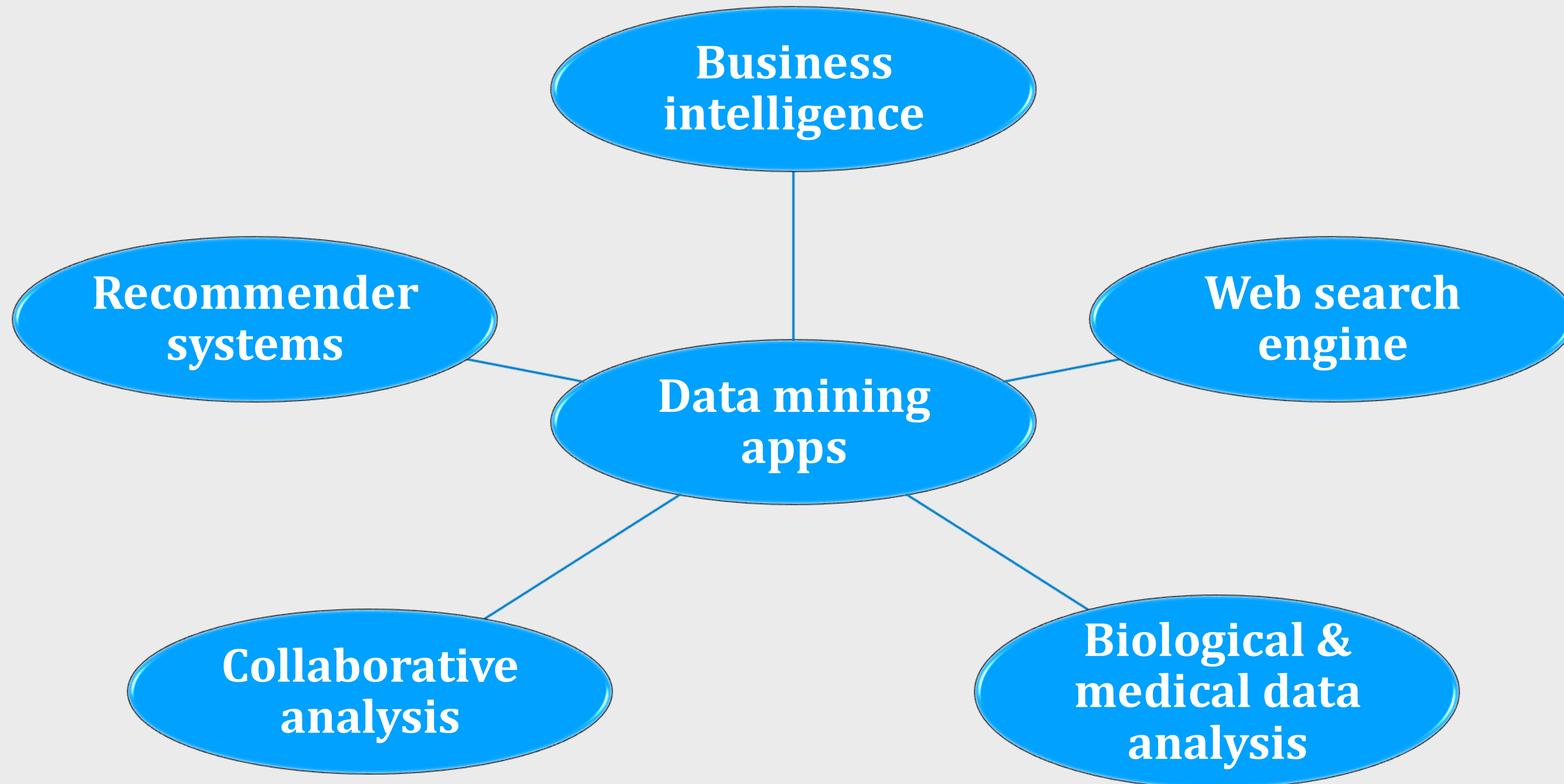
WHY CONFLUENCE OF MULTIPLE DISCIPLINES?

- Tremendous amount of data
 - Algorithms must be scalable to handle big data
- High-dimensionality of data
- High complexity of data
 - Data streams and sensor data
 - Time-series data, temporal data, sequence data
 - Structure data, graphs, social and information networks
 - Spatial, spatiotemporal, multimedia, text and Web data
 - Software programs, scientific simulations
- New and sophisticated applications

DATA MINING AND APPLICATION



DATA MINING AND APPLICATION





MAJOR ISSUES IN DATA MINING

- Handling noise, uncertainty, and incompleteness of data.
- Handling complex types of data
- Efficiency and scalability of data mining algorithms
- Privacy-preserving data mining

SUMMARY

- Data mining: Discovering interesting patterns and knowledge from massive amounts of data.
- A KDD process includes data preparation, data mining, pattern evaluation, and knowledge presentation.
- Data mining task: data summarization, mining frequent patterns, predictive analysis, cluster analysis, deep learning, outlier analysis.
- Data mining is a confluence of multiple disciplines.
- Data mining has broad applications.

EXERCISE

1. Which one is NOT an alias of data mining?
 - a. Knowledge discovery
 - b. Data analysis
 - c. Data replacement
 - d. Knowledge extraction

2. Describe one challenge that may arise during the data cleaning phase in the KDD process and how it can be addressed.

3. Present an example where data mining is crucial to the success of a business and mention what data mining task should perform.

REFERENCES

Han, J., Kamber, M., & Pei, J. (2023). Data Mining: Concepts and Techniques (4rd ed.). San Francisco, CA, USA: Morgan Kaufmann Publishers Inc.